

ECG Anomaly Detection — Session Report

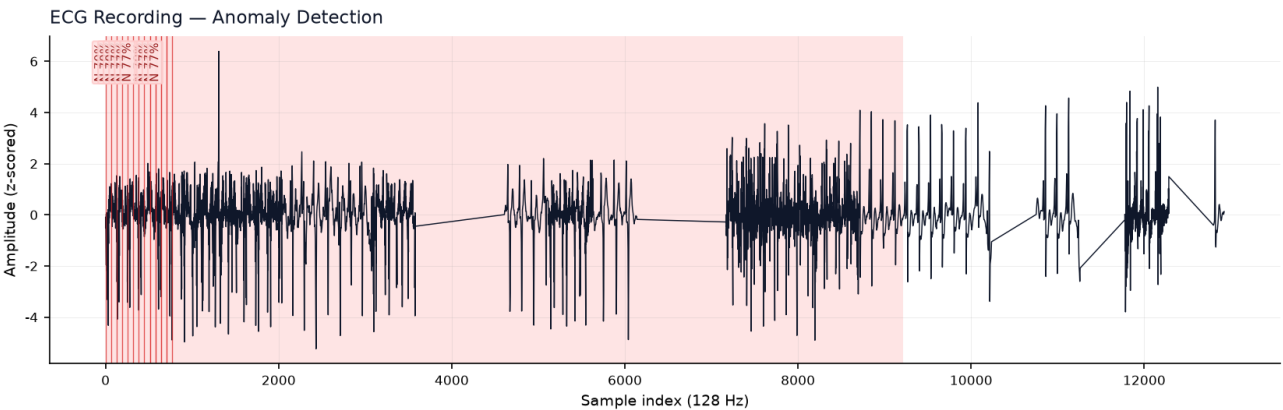
Prepared for: jj • Session ID: bc10fd03 • Generated: 2026-07-29 04:05:33

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (108)
Started	2026-07-29 04:05:33
Ended	2026-07-29 04:06:42
Duration	68.4 s
Status	completed
Total beats analyzed	34,304
Anomalous beats	22,175 (64.64%)
Model	ecg-ae-v2 (v1783189189) • Dilated-U-Net-Autoencoder
Detection threshold (MAE)	0.1341

ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red-shaded regions indicate samples flagged as anomalous by the autoencoder (reconstruction error exceeding the trained threshold).



Main Arrhythmia Finding

Premature Atrial Contraction was the main high-confidence arrhythmia type detected in this session. It appeared 122 time(s) with average confidence 69.2% and maximum confidence 86.8%.

Reason: this type is shown because the autoencoder first flagged the ECG segment as anomalous by reconstruction error, and the classifier then assigned this arrhythmia label with at least 50% confidence. Lower-confidence labels are excluded from this report to avoid over-interpreting uncertain model output.

Example: time 2026-07-29 04:06:15, anomaly score 0.1215, threshold 0.0568, classifier confidence 86.8%.

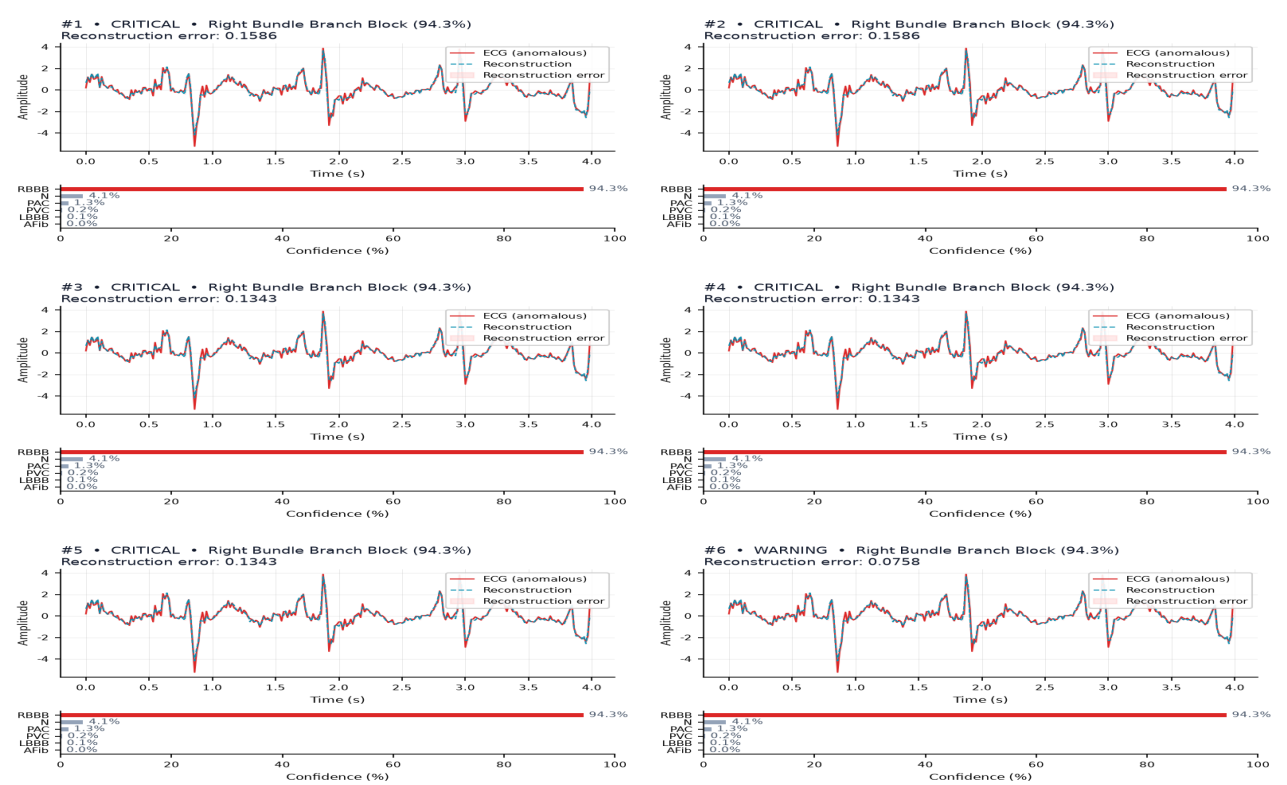
Clinical meaning: A Premature Atrial Contraction (PAC) is an early heartbeat that originates in the atria (the heart's upper chambers) instead of the SA node. Like PVCs, PACs can feel like a skipped beat or a brief fluttering sensation. PACs are generally less concerning than PVCs because they originate in the atria.

Why it matters: PACs are generally benign and very common, especially with aging. Most people experience occasional PACs without knowing it. In rare cases, very frequent PACs may be a precursor to atrial fibrillation. If you have frequent PACs, monitoring by a cardiologist is recommended.

Arrhythmia Type	Detections	Avg Confidence	Example
Premature Atrial Contraction	122	69.2%	score 0.1215, conf 86.8%
Right Bundle Branch Block	84	77.3%	score 0.1586, conf 94.3%
Premature Ventricular Contraction	23	59.9%	score 0.1140, conf 82.3%
Atrial Fibrillation	7	59.7%	score 0.1294, conf 59.7%

Detected Anomaly Waveforms

The following plots show example ECG segments (red) that triggered alerts, alongside the autoencoder's reconstruction of what it expected to see (blue dashed). The shaded region between the two curves represents the reconstruction error that exceeded the detection threshold. When available, examples are selected from arrhythmia classifications above 50% confidence.



Clinical Summary

64.6% of analyzed samples were flagged as anomalous, suggesting sustained abnormal ECG morphology. Urgent cardiology review is recommended. High-confidence classifier findings in this report include: Premature Atrial Contraction, Right Bundle Branch Block, Premature Ventricular Contraction,

Atrial Fibrillation. Only arrhythmia labels with confidence above 50% are included.

Physician Notes

Physician signature: _____

Date: _____

This report is generated automatically by an unsupervised deep-learning anomaly detection system and is intended for decision support only. It does not constitute a medical diagnosis and should not replace clinical judgment by a qualified physician.